

Module 2

1. Brief about Split MAC architecture and explain how it improves AP's performance.

Split MAC architecture is used in enterprise Wi-Fi networks where the functions of the Access Point (AP) are divided between the AP and a central controller (WLC – Wireless LAN Controller).

In this architecture:

- The AP handles lower-level tasks such as signal transmission, receiving data, encryption, and basic communication with clients.
- The controller handles higher-level tasks such as authentication, security policies, roaming, traffic management, and network control.

How it improves performance:

- Reduces load on AP since complex processing is moved to the controller
- Centralized management makes configuration and updates easier
- Improves scalability as many APs can be controlled together
- Enables better load balancing and traffic optimization
- Supports faster roaming between APs
- Improves overall network efficiency and performance

2. Describe the CAPWAP, explain the flow between AP and controller.

CAPWAP stands for Control and Provisioning of Wireless Access Points. It is a protocol used for communication between Access Points (APs) and the Wireless LAN Controller (WLC) in enterprise Wi-Fi networks.

CAPWAP uses two tunnels:

- Control tunnel for management and configuration (encrypted)
- Data tunnel for user data traffic between AP and controller

Flow between AP and Controller:

1. AP starts and gets an IP address using DHCP
2. AP discovers the controller (WLC) on the network
3. AP establishes a secure CAPWAP connection with the controller
4. Controller sends configuration details to the AP
5. AP starts broadcasting SSIDs and allows client connections
6. Client data is sent from AP to controller through CAPWAP tunnel
7. Controller manages security, traffic, and network policies

3. Where the CAPWAP fits in the OSI model, what is the two tunnels in CAPWAP and its purpose.

CAPWAP operates over the Transport Layer (Layer 4) using UDP, and functions across Layer 3 (Network Layer) as it runs over IP networks. It is used to carry wireless data and control information between the Access Point (AP) and the Wireless LAN Controller (WLC).

CAPWAP uses two tunnels:

1. **Control Tunnel**
Used for communication between the AP and controller for configuration and management. It carries messages such as AP registration, authentication, and configuration updates. This tunnel is encrypted for security.
2. **Data Tunnel**
Used to carry user data traffic from wireless clients. It forwards data from the AP to the controller for processing and routing.

4. What is the difference between Lightweight APs and cloud-based APs ?

Feature	Lightweight APs	Cloud-based APs
Controller Location	On-premise (local WLC)	In the cloud
Management	Centralized via local controller	Centralized via cloud dashboard
Data Traffic	Usually passes through controller	Data stays local (not sent to cloud)
Setup	Requires controller hardware	No physical controller needed
Deployment	Used in large single-location networks	Used in distributed/multi-location networks
Cost	Higher initial cost (hardware)	Lower hardware cost, subscription-based
Access	Managed within local network	Managed remotely from anywhere
Scalability	Limited by controller capacity	Easily scalable through cloud

5. How is the CAPWAP tunnel maintained between AP and controller?

- The AP first gets an IP address using DHCP
- It discovers the Wireless LAN Controller (WLC)
- A CAPWAP session is established between AP and controller
- Two tunnels are created: control tunnel and data tunnel
- The control tunnel is continuously used to exchange keep-alive (heartbeat) messages
- These messages ensure that the AP and controller are still connected
- If the connection is lost, the AP tries to reconnect to the controller

6. What is the difference between sniffer and monitor mode, use case of each mode.

Feature	Sniffer Mode	Monitor Mode
Purpose	Captures wireless packets	Monitors wireless network activity
Function	Sends captured packets to controller for analysis	Scans network for issues and rogue devices
Client Service	Does not serve clients	Does not serve clients
Data Handling	Focuses on packet capture	Focuses on network monitoring and reporting
Use Case	Troubleshooting, debugging, packet analysis (Wireshark)	Detecting rogue APs, interference, and network health monitoring

7. If WLC depends on WAN, which AP mode is best for the local network and how ?

The best AP mode is FlexConnect mode.

- In FlexConnect mode, the AP connects to the central WLC but can handle traffic locally
- Client data is not sent through the WAN, it is switched directly to the local network
- Even if the WAN connection fails, the AP can continue to serve connected clients
- New clients can also authenticate locally
- It reduces latency and improves performance for applications like voice and video

8. What are the challenges if deploying autonomous APs (more than 50) in large network like a university?

- **No centralized management:** Each AP must be configured manually, which is time-consuming
- **Difficult to scale:** Managing 50+ APs becomes complex and inefficient
- **Interference issues:** APs cannot coordinate channels, leading to signal interference
- **Poor roaming experience:** Devices may disconnect or experience delay when moving between APs
- **Security limitations:** No central monitoring for threats or rogue access points
- **High maintenance effort:** Updates and troubleshooting must be done individually on each AP

9. What happens on a wireless client connected to Lightweight AP in local mode if WLC goes down?

- All connected clients lose network connectivity because the AP depends on the WLC to forward data
- Existing clients may still remain connected to the Wi-Fi (SSID), but no internet or data transfer will work
- New clients cannot connect or authenticate since the WLC handles authentication
- The AP continues to broadcast the SSID, but it cannot process or route traffic
- Since encryption keys and policies are managed by the WLC, communication stops when it is unavailable
- The AP continuously tries to reconnect to the primary or backup WLC